IoT-Based LPG Vehicle Tracking and Safety System

The proposed platform consists of an ESP32 controller, an MQ-6 sensor for LPG leak detection, an HC-SR04 ultrasonic sensor for **proximity hazard identification**, a Neo-6M GPS module for **live tracking**, and SIM900A GSM or Wi-Fi modules for **mobile alerts.** On event detection, **the system activates local signals, a buzzer, and an LED** and **instantly escalates notifications to preconfigured contacts and remote authorities**. All sensor and event data is seamlessly **ingested into a cloud database for post-incident analysis**, maintenance scheduling, and compliance review. The unified approach ensures **scalable deployment for both household cylinder delivery and bulk transport**, substantially improving response efficiency and reducing operational risks

Components

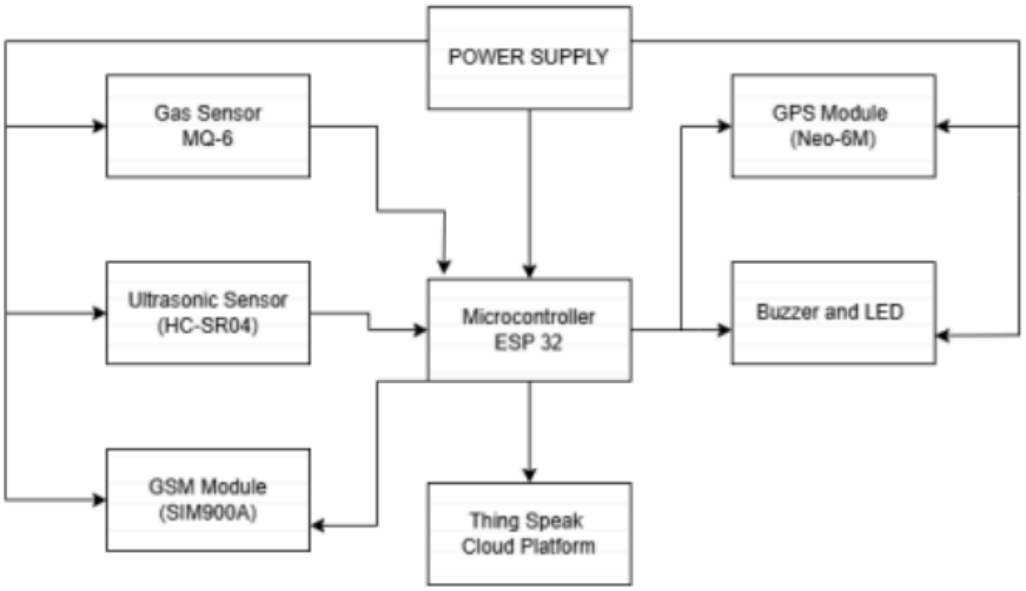
Hardware

- ESP32 Microcontroller
- MQ-6 Gas sensor
- Ultrasonic sensor HC-SR04
- GPS module Neo-6M
- GSM module SIM900A

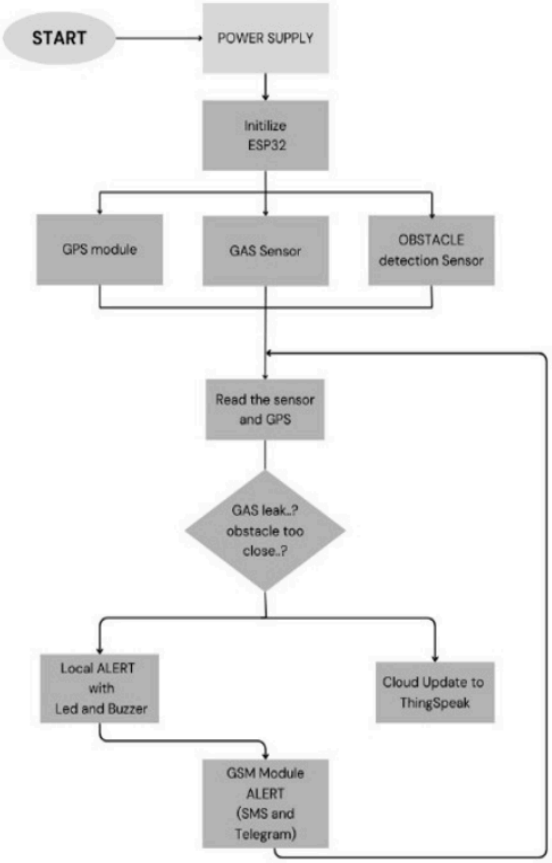
Software

- Arduino IDE
- Thingspeak
- Embedded C programming
- Telegram Bot API

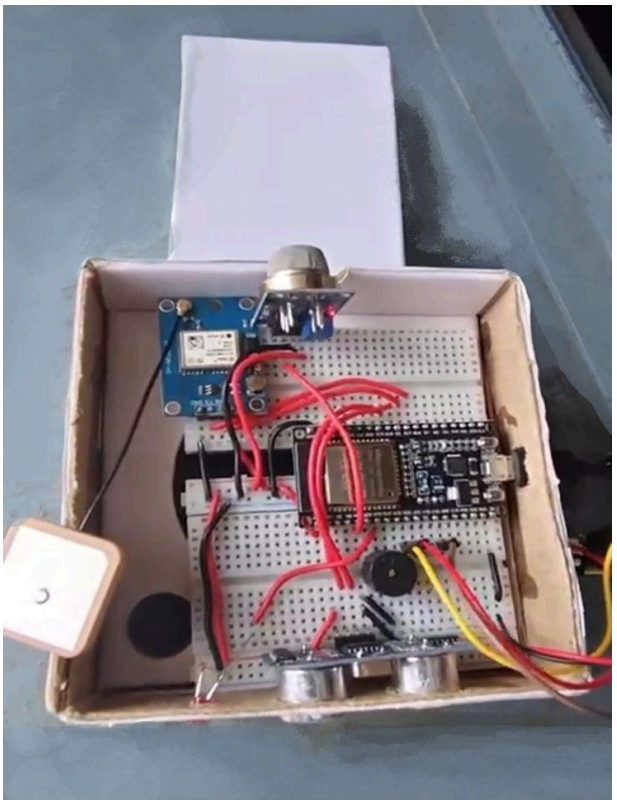
Block Diagram



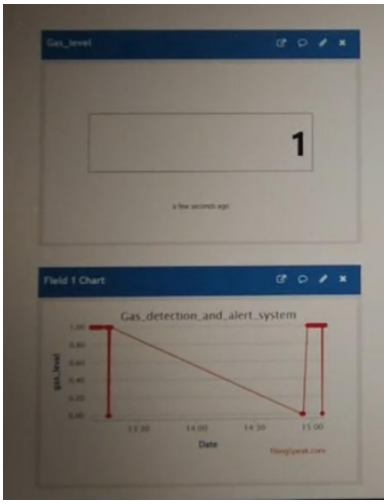
Methodology

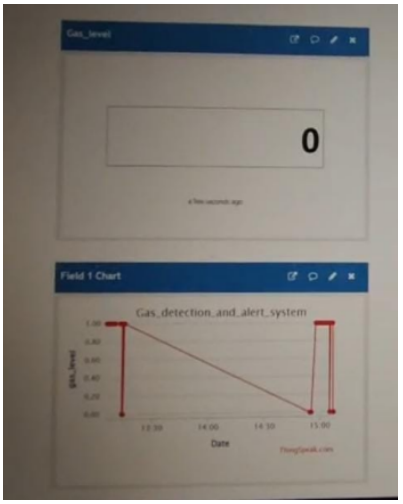


LPG Vehicle Tracking System

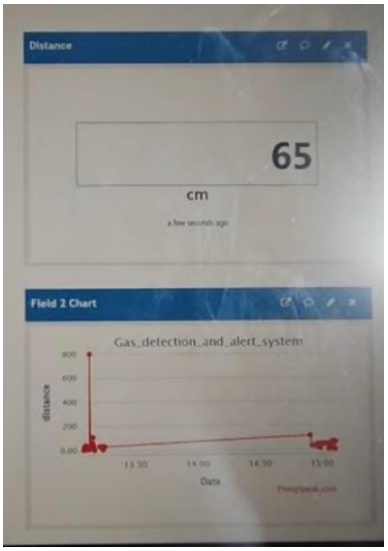


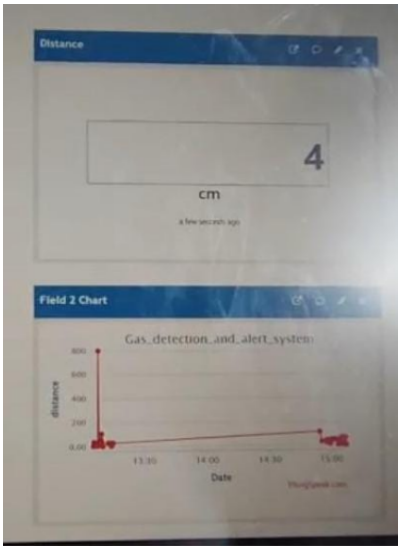
Graph



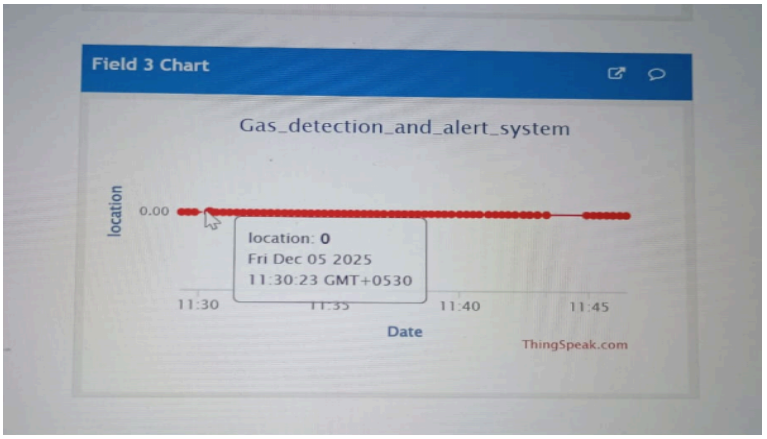


Concentration vs Time





Distance vs Time



Latitude and Longitude visualization